24CS2506 – Building Applications using GPT-4

EXPERIMENT 5 – AI-Powered Medical Symptom Assistant using GPT-4

PART 1 – PREPARATION MATERIAL

What is a Medical Assistant?

A **medical assistant application** is an AI-based system that helps users understand health-related information based on the symptoms and information they provide.

An AI medical assistant may help with:

- Symptom analysis
- Health information
- Possible conditions
- Follow-up questions
- General self-care information
- Identifying situations requiring professional medical attention

What is Symptom Analysis?

Symptom analysis is the process of examining the symptoms reported by a user and identifying possible conditions that may be associated with those symptoms.

Example

User Input:

I have a sore throat, mild fever and cough.

The system may respond:

```
Possible explanations may include:
• Common cold
• Influenza
• Other respiratory infections

These symptoms alone are not sufficient to establish
a diagnosis.
```

The system should communicate uncertainty rather than presenting one condition as a confirmed diagnosis.

Why Use GPT-4 for Symptom Analysis?

GPT-4 can understand natural-language descriptions.

A user does not have to enter symptoms in a fixed format.

Traditional Input

```
Fever = Yes
Cough = Yes
Headache = No
```

Natural Language Input

```
I've had a fever and cough since yesterday.
I also feel tired and have a mild headache.
```

An LLM can interpret the natural-language description and organize the information.

GPT-4 also has documented safety limitations, including the possibility of inaccurate information, so medical applications require additional safeguards and careful evaluation.

What Information Should the System Collect?

A symptom assistant should ideally collect relevant context before generating an analysis.

Basic Information

- Age
- Sex, where relevant
- Symptoms
- Duration
- Severity
- Previous medical conditions
- Current medications, if relevant
- Allergies, if relevant

Example

```
Age       : 22
Symptoms  : Fever, cough, sore throat
Duration  : 2 days
Severity  : Mild
```

Structured Output

The assistant should generate a structured response.

Example

```
MEDICAL SYMPTOM ANALYSIS
--------------------------------

Reported Symptoms:
• Fever
• Cough
```

```
• Sore throat

Duration:
2 days

Possible Conditions:
1. Common cold
2. Influenza
3. Other respiratory infection

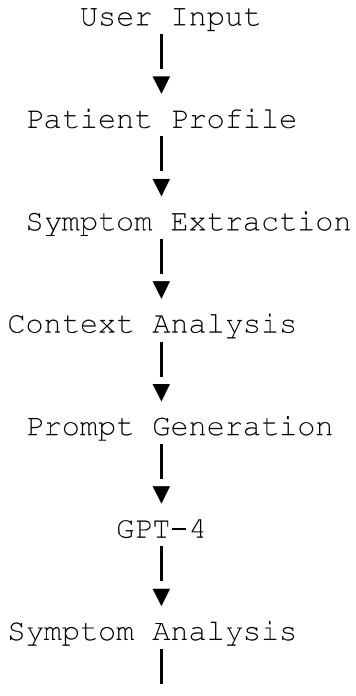
Why These Were Considered:
The reported symptoms can occur with several
respiratory conditions.

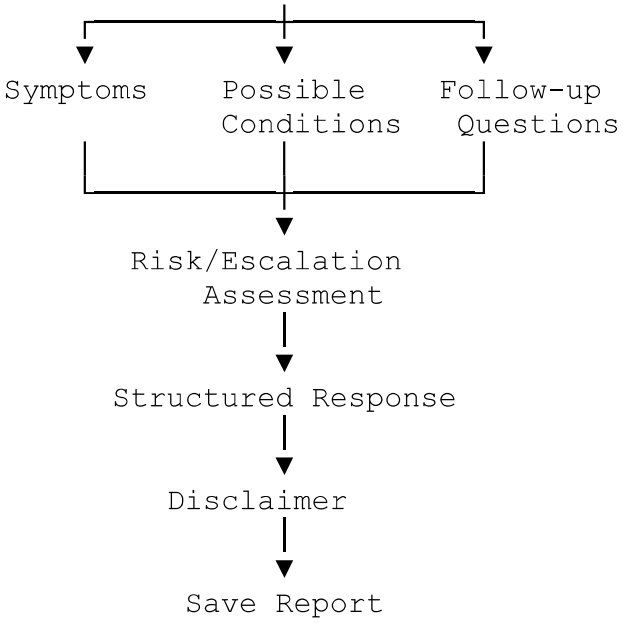
Additional Information Needed:
• Exact temperature
• Severity of symptoms
• Breathing difficulty
• Other associated symptoms

Medical Attention:
Seek professional medical evaluation if symptoms
are severe, worsening or concerning.

Disclaimer:
This is an informational AI analysis and not
a medical diagnosis.
```

System Workflow




Symptoms
Possible Conditions
Follow-up Questions
Risk/Escalation Assessment
Structured Response
Disclaimer
Save Report